

Evaluating NeRF Fidelity using Virtual Environments

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ABSTRACT

Neural Radiance Fields (NeRF) are a promising form of 3D Reconstruction that utilize sparse 2D imagery to recreate synthetic and real scenes. Since NeRF was first developed, numerous methods and techniques have worked upon the original algorithm to improve its accuracy, fidelity, and speed. We aim to evaluate the fidelity of a reconstructed scene, as well as evaluate the quality of the reconstruction.

Keywords: Augmented Reality, Visualization, 3D Reconstruction, NeRF

Index Terms: Computing methodologies - Artificial intelligence - Computer vision - Computer vision problems - Reconstruction; Human-centered computing - Visualization - Visualization application domains - Visual analytics; Computing methodologies - Artificial intelligence - Computer vision - Image and video acquisition - Camera calibration; Computing methodologies - Modeling and simulation - Simulation types and techniques - Real-time simulation;

1 INTRODUCTION

Creating virtual environments that accurately mimic the scene they are based on is a difficult task. Existing methods include laser scans, LIDAR scans, and creating models and rooms with tools such as Unity or Unreal Engine. Recreating a scene, synthetic or real, normally requires special equipment or experience in 3D modeling. Neural Radiance Fields (NeRF), [1] is a CNN approach to reconstructing scenes using only a camera.

NeRF implementations, on average, utilize the same pipeline. Using a camera, someone can take photos or a video of a scene from multiple viewpoints, upload the imagery into the algorithm, and end with a 3D visualization of the scene. This 3D visualization can then be turned into a point cloud, a mesh, or viewed natively in supported viewers.

2 GOALS AND INTENT

Our goals are to evaluate how closely a scene visually resembles its ground truth input, evaluate NeRF's capabilities in distance interpretation, and evaluate the difficulty of telling differences in a

scene that has been through a NeRF compared to one that has been modeled using a conventional method. The primary interest of these evaluations is to determine how close a NeRF's output is to its input.

2.1 Visual Fidelity

The primary methods of evaluating a NeRF's fidelity in comparison to its input include PSNR, SSIM, and LPIPS, [2] amongst a few others. Our focus on determining how well a scene was reconstructed will include metrics such as these but will mainly be regarding human interpretation. Once a participant has put the headset on, we will measure their response to the ground truth environment and compare it to their response in the NeRF environment.



Figure 1: Three images from the same view. From left to right: Unity view, pixel subtraction view, NeRF view.

We also intend to use a basic pixel subtraction method (as seen in Figure 1) to see where the highest points of difference can be found. Preliminary testing shows that most differences in both synthetic and real-world scenes can be found on edges of objects, in areas with intricately detailed textures, areas with poor lighting, and areas without proper viewpoints from the camera's inputs.

2.2 Distance Interpretation

Visual similarities from specific viewpoints will help determine how accurate a scene is from its ground truth, but we also want to determine how accurate the environment is to the space it is occupying. One of our major goals to determine this is to measure distances from various points in the NeRF and compare them to the same points in the ground truth.

Scaling a NeRF to its ground truth size can be unreliable, as objects in the NeRF are not always fully reconstructed to scale, nor are they always completely accurate.

3 METHODOLOGY

All NeRF renders used will be created in the Hi5 Lab using higher-end consumer level graphics cards, such as the RTX 4080 and 4090. Although these GPUs are not necessary in the creation of accurate reconstructions using NeRF, the more power a GPU has, the faster the NeRF generates. Human subject testing will involve

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participants from the Hi5 Lab, as well as others from the CSE department at Mississippi State University.

All real-world captures will be made using a second-generation iPhone SE, and all synthetic captures will be made using Unity. All virtual environment testing will be done with the Oculus Quest 3.

3.1 Virtual Environments and NeRF Implementations

To test the NeRF using virtual environments, we will export their resulting meshes into Unity. NeRF meshes tend to have less fidelity compared to their visualizations, although this can depend on the chosen NeRF implementation. For our current experiments, we are utilizing nerfstudio's nerfacto [3] methods and Instant-NGP's [4] VR integration.

Currently, the most straightforward way to view a NeRF visualization in a VR headset without first pulling it down as a mesh or point cloud is to use Instant-NGP's NeRF implementation. We are working on adding a method to nerfstudio to view visualizations in a VR headset.

3.2 Human Testing

There are several methods of evaluating visual fidelity and accuracy of a NeRF reconstruction. Most of these metrics are based on pixel-space comparisons. We aim to evaluate accuracy and fidelity of the NeRF reconstruction through response of participants viewing the environments we create.

Currently, we have determined that more work will need to be done before we can use the meshes generated by the average NeRF implementation. Although the meshes generated by nerfacto are relatively high quality, they tend to have holes in areas of low confidence. The meshes tend to produce lower quality results when attempting to render an entire scene rather than an object.

As the meshes are currently too low quality to justify running survey tests on, we will instead use the visualizations produced by Instant-NGP, as the VR module allows for participants to view the NeRF in a headset.

4 FUTURE WORK

We are hopeful that NeRF reconstructions will have a high enough visual fidelity that they will be applicable to creating virtual environments that mimic the real world. Once testing has been completed in determining the fidelity of NeRF virtual environments, we would like to run human studies on scene recall. We would like to NeRF real world objects and place them in virtual environments, synthetic, real, or a hybrid of the two, and ask users to tell the difference between a real-world scene and the altered real-world scene that contains the NeRF object.

We are also optimistic that NeRF can produce accurate distance measurements and would like to run further studies on how accurate NeRF is at measuring larger distances, such as distances upwards of twenty meters, as well as smaller distances downwards of one centimeter. Assuming that NeRF can create spaces with accurate distances between points, we would like to consider creating methods to measure distances empirically using NeRF instead of relying on human tests.

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